CS396 Project Phase 2 Report

epaData: Analysis, Impact Assessment, and Prediction for EPA Power-Sector Emissions Data

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Abstract

Phase 1 of epaData built a database of the annual operating and emissions data that U.S. power-plant units report to the EPA Clean Air Markets Program (CAMPD), with retrieval, upload, validation, search, and export. Phase 2 turns that database into a decision-support tool. An *Analyze* page lets a user select unit-years with the Phase 1 filters and then summarize them with descriptive statistics, derive normalized indicators such as emission intensities, heat rate, capacity factor, and year-over-year change, explore them in interactive charts, compare groups with confidence intervals and hypothesis tests, and convert CO_2, SO_2, and NO_x emissions into EPA TRACI 2.2 potential-impact indicators. A *Predict* page compares a baseline with Ridge regression and a decision tree that predict a unit's annual NO_x emissions; on the held-out reporting years 2024–2025 the tree reaches $R^2 = 0.951$ against 0.489 for the baseline. The same models drive one-variable sensitivity curves and named what-if scenarios, with warnings when an input leaves the training data. Every result can be exported with metadata that records how it was produced. This report documents the analytical questions, the methods and their assumptions, the design, the results, and the limitations of the system.

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1 Project description

1.1 Problem

Phase 1 made CAMPD data easy to find and trust: a user can ask which Kentucky coal units emitted more than 500,000 tons of CO_2 in 2025 and see where every number came from. It could not answer the questions that follow. Is the sector's CO_2 falling because plants are getting cleaner or because generation is moving between fuels? Do units with selective catalytic reduction (SCR) really emit less NO_x per unit of fuel, and by how much? Which states contribute most to acidification, and does the ranking change once the result is divided by the electricity produced? How much NO_x would a particular unit emit if it burned 5 % less fuel or added a control?

Answering such questions requires statistics over many unit-years with a correct treatment of missing values, indicators that are comparable across plants of different size, a defensible link between emitted mass and environmental impact, and a model whose predictions are evaluated honestly before they are used for what-if analysis.

1.2 Analytical questions

Phase 2 is organized around four questions, each answered with a function of the application (Section 5):

Q1. How have emissions and emission intensity changed between 2015 and 2025, and how much of the change is a shift from coal to gas?

Q2. How large is the difference in NO_x emission rate between units with and without post-combustion NO_x control, and is it statistically distinguishable from chance?

Q3. Which states and facilities contribute most to each TRACI impact category, in total and per MWh?

Q4. How well can a unit's annual NO_x emissions be predicted from its operating data, attributes, controls, and history, and how do predictions respond to changed inputs?

1.3 Intended users

The specification names environmental researchers, power-sector analysts, regulators, facility managers, students, and the interested public. Phase 2 assumes these users know their domain but not statistics software: every statistic is labeled with its unit and definition, every test is accompanied by a plain-language reading, and every model result is labeled as a model response rather than a measured effect.

1.4 Objectives

Phase 2 addresses the course objectives deferred from Phase 1, O8 (environmental indicators), O9 (multi-criteria evaluation), and O10 (interactive charts of evaluation results), through the nine goals of the Phase 2 specification: explore the data with descriptive statistics; compare facilities, units, fuels, states, and periods; visualize emissions and operating data; estimate potential impacts with TRACI factors; identify trends, relationships, and unusual observations; build and evaluate prediction models; perform sensitivity and scenario analysis; save and export results; and explain methods, assumptions, results, and limitations. Section 6 assesses each goal.

1.5 Scope

Phase 2 uses the CAMPD records of the Phase 1 database and the TRACI 2.2 characterization factors for the three pollutants CAMPD reports (CO_2, SO_2, and NO_x emitted to air). Analyses default to the complete reporting years 2015–2025 retrieved from the CAMPD API; the older uploaded years (1996–1998 and 2006–2008) and the partial year 2026 remain selectable but are excluded from model training. The prediction target is annual NO_x mass per unit-year (Section 4.5). Multi-criteria weighted scoring, listed among the course objectives, is not part of the Phase 2 specification and was not built.

2 Functions implemented

Two pages were added between *Map* and *Retrieve* in the navigation bar: *Analyze* (`/analyze`), with the tabs Dataset, Statistics, Charts, Compare, and Impacts, and *Predict* (`/predict`), with the tabs Models, Sensitivity, and Scenarios. As in Phase 1, the URL holds the complete state of a page (filters, tab, and every choice on the tab), so any view can be reloaded, bookmarked, or shared.

2.1 Analytical dataset builder

The Analyze page reuses the Explore page's filter bar without change: reporting-year range, state, facility, unit, primary and secondary fuel, unit type, SO_2, NO_x, and particulate controls, program, operating status, record origin, audit flags, and minimum and maximum operating time, gross load, heat input, and emitted mass. The selected unit-years are the analytical dataset for every tab. The Dataset tab (Figure 1) reports how many unit-years, facilities, units, and years were selected, lists every variable with its kind, unit, definition, how groups combine it, and how many of its values are missing, and lists the fields the selection can be grouped by with their number of distinct values. A *Download dataset* button exports every selected unit-year with all variables (Section 2.9).

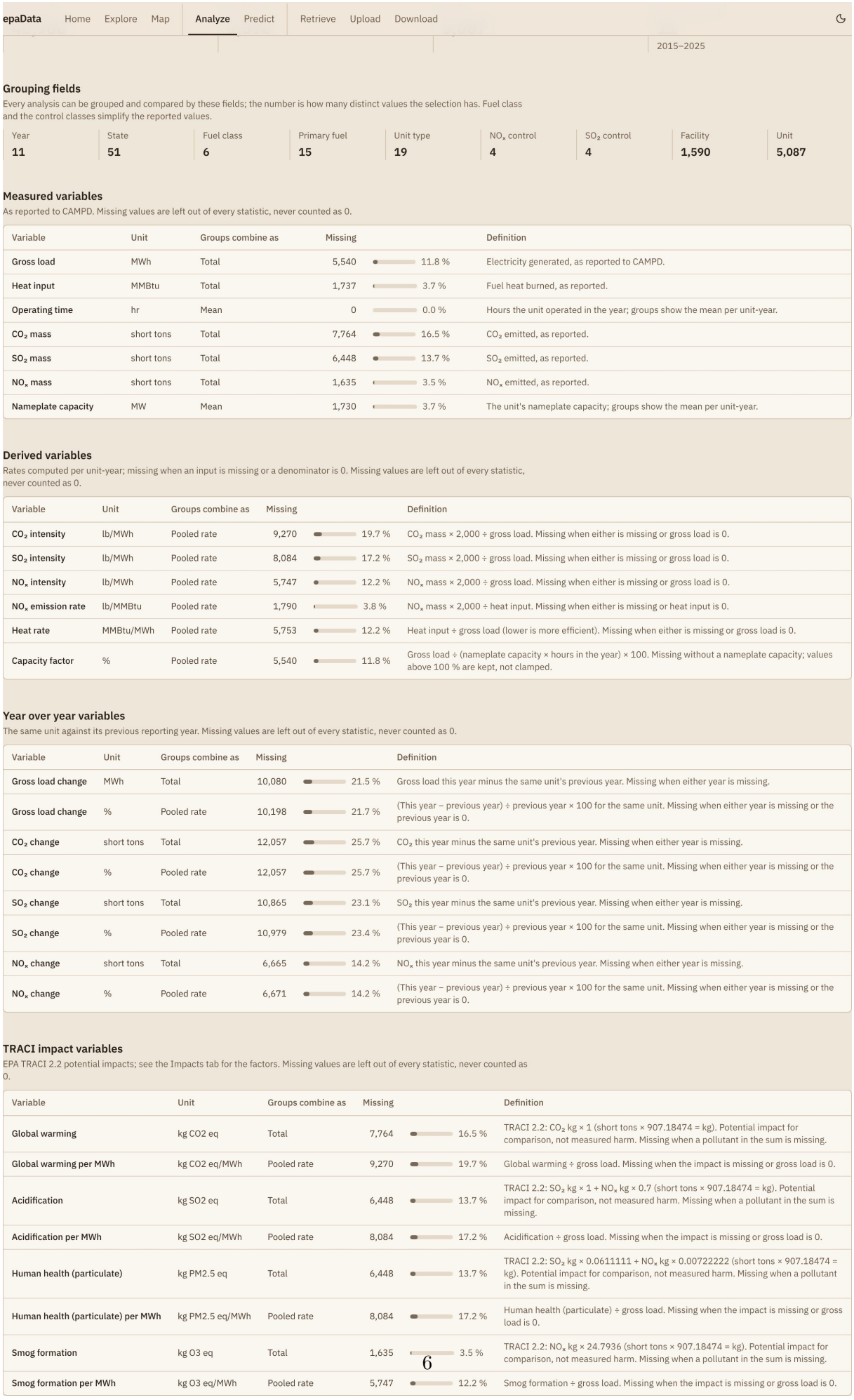


Figure 1: Dataset tab for 2015–2025: 46,988 unit-years of 1,590 facilities and 5,087 units. Each variable is listed with its kind, unit, how groups combine it, and its missing values.

2.2 Derived indicators

Twenty-nine variables are available (Table 3): seven measured values, six derived rates, eight year-over-year changes, and eight TRACI indicators. The derived and TRACI variables are computed from the stored records whenever an analysis runs, so they always agree with the current data. Section 4 gives their definitions and the rules for missing values.

2.3 Descriptive statistics

The Statistics tab (Figure 2) describes one variable over the selection: count, missing count, minimum, maximum, mean, median, standard deviation, first and third quartiles, interquartile range, and the 5th, 10th, 90th, and 95th percentiles, together with the variable's total, mean, or pooled rate over the whole selection. With a grouping field it repeats these statistics for each group in a sortable table. It also lists *unusual observations*: the unit-years beyond Tukey's far-out fences, $Q_1 - 3\,\mathrm{IQR}$ and $Q_3 + 3\,\mathrm{IQR}$, with their count on each side and the ten most extreme. The four example questions of the specification (average CO_2 by state, median NO_x intensity by fuel, total SO_2 by year, and average heat rate by unit type) are offered as one-click examples.

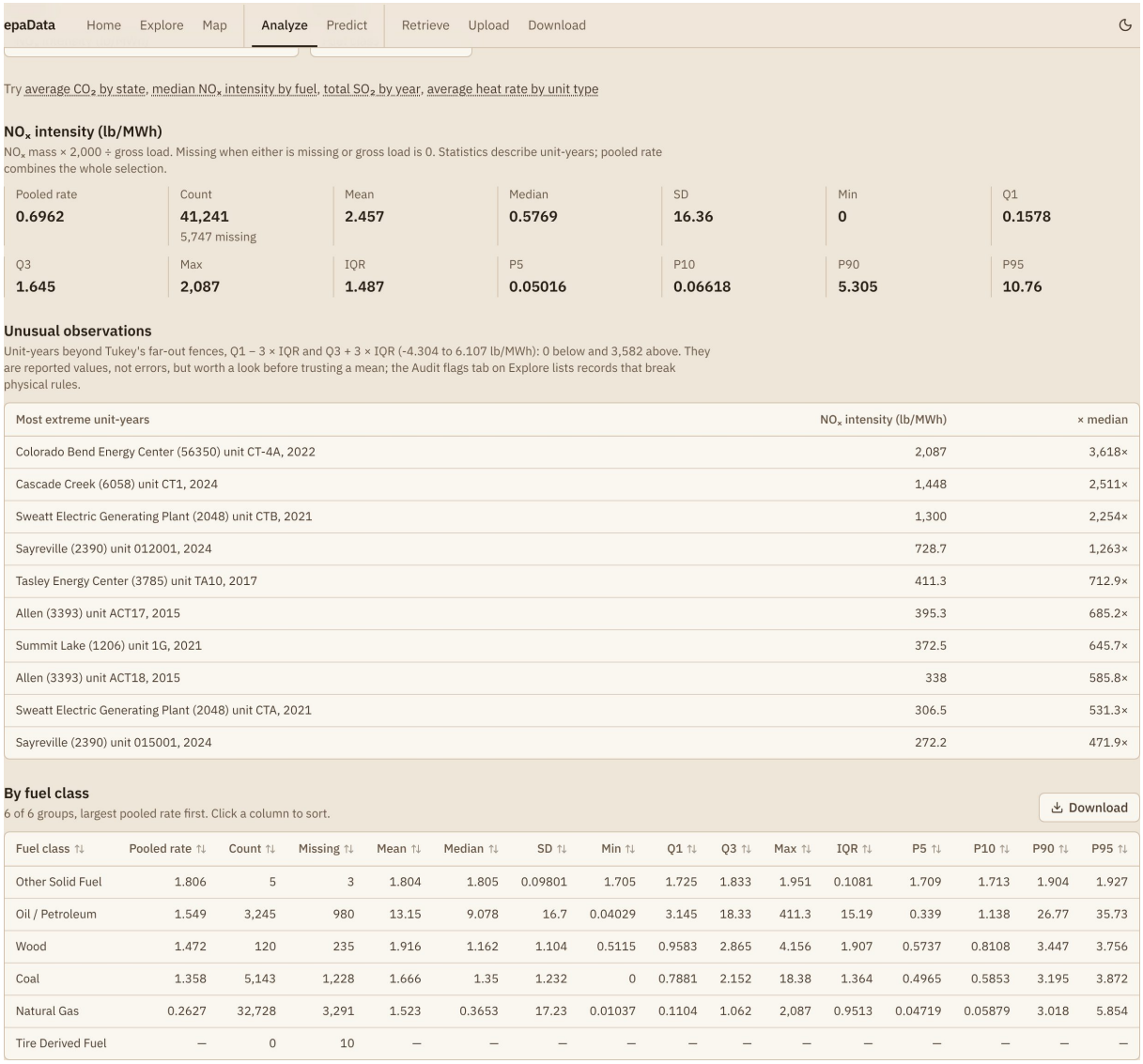


Figure 2: Statistics tab: NO_x intensity over 2015–2025, overall and by fuel class.

2.4 Interactive visualization

The Charts tab draws seven chart types with Plotly.js [8], all over the same selection: a trend of any variable per year with one line per group; bars per group; histograms and box plots, optionally on a logarithmic scale and split by group; scatter plots of any two variables colored by group; a Pearson or Spearman correlation heat map of eleven variables; the highest or lowest twenty facilities or units; and a state choropleth. Every chart has a descriptive title, axis titles with units, a legend, tooltips with values and counts, zoom, PNG export, and a *Data* button that saves its data, image, and metadata. Fuel classes keep the colors they have on the Phase 1 map. Distributions and scatter plots of very large selections are drawn from an even sample of 20,000 values or 4,000 points, which the chart states; correlations and statistics always use every value.

2.5 Comparative analysis

The Compare tab compares two to eight groups along any grouping field: states, fuels, unit types, NO_x or SO_2 control classes (for example units with and without SCR), or reporting years

(for example 2015 against 2024). For each group it shows the pooled value of the variable, its rank among the compared groups, its difference and percentage difference from the first group, the count, the mean with its 95 % confidence interval, the median, and the standard deviation; it runs Welch's t-test and the Mann–Whitney U test for two groups and one-way ANOVA and the Kruskal–Wallis test for any number; and it states the result in plain language with its caveats (Figure 6). The Phase 1 comparison of two to four individual facilities or units remains available from the Explore page.

2.6 Environmental and health impact indicators

The Impacts tab converts the selected emissions into four TRACI 2.2 impact categories, global warming, acidification, particulate-matter human health, and smog formation, grouped by any field. Each category is shown in its own equivalent unit, both as a total and per MWh of gross load, and is never added to another category. The tab shows the factors used, with their TRACI version, substance, CAS number, and unit, and the formula of each category, and it states that the results are potential impacts for comparison, not measured harm (Figure 8).

2.7 Prediction

The Models tab of the Predict page compares the three registered models, a baseline, Ridge regression, and a decision tree, on the validation and test years; plots predicted against actual NO_x for the test years; shows the tree's feature importances or the Ridge coefficients; and records each model's inputs, split, saved file, software versions, and training time (Figure 9).

2.8 Sensitivity and scenario analysis

The Sensitivity and Scenarios tabs start from a baseline unit-year, chosen by facility name or ID, year, and unit, and a model. Sensitivity varies one input from a minimum to a maximum value, entered either in the input's own unit or as percentages of its baseline value, with all other inputs held constant, and draws the prediction as a curve with the baseline marked and the region outside the training data shaded (Figure 10). Scenarios compares the baseline with five named changes, gross load +10 %, heat input −5 % (both editable), adding or removing SCR, switching the primary fuel, and a combined high-efficiency case, for both trained models side by side (Figure 11).

2.9 Saving and exporting results

Every chart's *Data* button and every statistics and impact table downloads a ZIP file containing the data as CSV, the chart as PNG when there is one, and a `metadata.json` file. The metadata records the export time and URL; the software version and Git commit; a snapshot of the database (unit-years, datasets, and last import); the filters and every tab choice; the definition, unit, and aggregation of each variable shown; the TRACI factors for impact results; and, on the Predict page, the model's inputs, settings, split years, metrics, saved file, and software versions, and the baseline unit-year. The complete analytical dataset is exported by the server (`/api/export?type=analysis`) in the same format.

2.10 Requirement traceability

Table 1 maps each requirement of the Phase 2 specification to the function that meets it.

Table 1: Phase 2 requirements and where they are met.

Specification	epaData	Status
§3 CAMPD multi-year data	2015–2025 API years (and older uploads) in `annual_records`	Met
§3 TRACI version, pollutant, category, factor, unit stored	`traci_factors` table	Met
§4.1 Dataset builder: years, state, facility, unit, fuel, unit type, controls, program, minimum operating time or load	Explore filter bar on Analyze	Met
§4.1 Record, facility, unit, year counts; missing-value summary; variables	Dataset tab	Met
§4.1 Reproducible through URL or saved configuration	URL parameters; exported metadata	Met (URL)
§4.2 Count, missing, min, max, mean, median, SD, quartiles, IQR, percentiles; grouped	Statistics tab	Met
§4.3 CO_2, SO_2, NO_x intensity; heat rate; year-over-year change; utilization	Derived variables; capacity factor	Met
§4.3 Division by zero and missing values stay missing	Variable rules; tests	Met
§4.4 Line, bar, histogram, box, scatter, correlation heat map, rankings, state map	Charts tab	Met
§4.4 Titles, units, legends, tooltips, filters, image and data download	`Chart` component	Met
§4.5 Facilities, units, states, fuels, unit types, with and without control, periods	Compare tab; Phase 1 compare dialog	Met
§4.5 Statistics, differences, rankings, charts, normalized indicators; tests; plain language	Compare tab	Met
§4.6 Short tons to kg; exact pollutant matching; sum within category; per MWh; record factors	Impacts tab; `impactVariables`	Met
§5.1 One prediction target	Annual NO_x mass per unit-year	Met
§5.2 Features; no target leakage	Section 4.5; ablation	Met
§5.3 Baseline and at least two models	Median baseline, Ridge, decision tree	Met
§5.4 Chronological split; MAE, RMSE, R^2, MAPE	2016–21 / 2022–23 / 2024–25	Met
§6.2 One variable at a time	Sensitivity tab	Met
§6.3 Scenarios with range warnings	Scenarios tab	Met

Specification	epaData	Status
§6.4 Curve, baseline marks, scenario bars, tooltips, out-of-range shading	Sensitivity and Scenarios tabs	Met
§7 Export datasets, statistics, models, sensitivity, scenarios, charts, with metadata	ZIP exports	Met
§8.1 Flask, Jinja2, SQLAlchemy, pandas, scikit-learn, joblib, pytest	Next.js, React, Drizzle; pandas, scikit-learn, joblib, pytest offline (Section 3)	Changed
§12.3 Instructions, migrations, tests, fixtures, no keys	README, migration 0009, 57 + 6 tests	Met

2.11 Planned functions not completed or changed

Table 2 compares the Phase 2 plan with the delivered system.

Table 2: Plan versus delivered system.

Planned	Delivered	Reason
Store derived rates as columns	Computed when an analysis runs	Always consistent with the records; no backfill after imports.
Eutrophication from NO_x	Not provided	The TRACI 2.2 workbook has no eutrophication column.
Partial impact when a pollutant is missing	Category missing for that unit-year	A partial sum would understate the impact silently.
State and capacity factor as model inputs	Left out	State is sparse for many fuel and type combinations; capacity factor duplicates gross load and nameplate capacity.
Saved analysis configurations	URL parameters only	The URL already reproduces a view; saving it is a bookmark.
Gradient-boosting model for comparison	Not built	Time; the tree already reaches $R^2 = 0.95$.

3 System design

3.1 Architecture

Phase 2 keeps the Phase 1 stack: Next.js 16 with React 19 on the web layer, tRPC procedures for typed calls from the browser, Drizzle ORM, and SQLite. The specification recommends Flask with pandas and scikit-learn; the Phase 1 report explained the mapping of Flask, Jinja, and SQLAlchemy to the stack used. For the analytical work we adopted a hybrid (Figure 3). Statistics, indicators, and TRACI impacts are computed in TypeScript on the server, close to the database. Models are trained offline in Python with pandas and scikit-learn [7], and each trained model is stored in the database with its preprocessing and parameters as JSON, which a small TypeScript module replays exactly. Predictions, sensitivity curves, and scenarios are therefore instant, need no Python process at request time, and work on the read-only hosted build, where Python is unavailable.

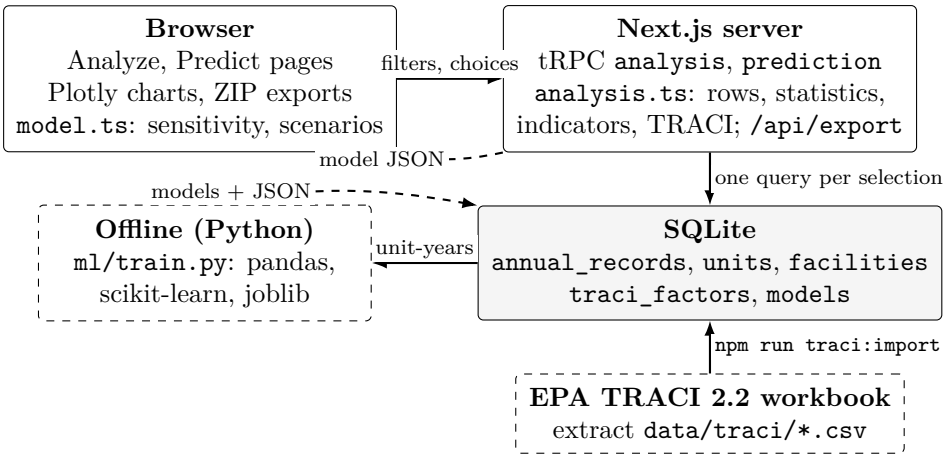


Figure 3: Phase 2 architecture. Solid arrows run on every request or import; dashed arrows carry the trained models from the offline pipeline to the browser.

3.2 Life cycle of an analysis

When a tab needs a result, the browser calls a tRPC procedure with the filters and the tab's choices. The server turns the filters into the same SQL conditions the Explore page uses (one shared function, `unitYearConditions`), and selects the matching unit-years joined with their unit, facility, and the same unit's previous year, which the year-over-year variables need. The rows are cached for 60 seconds, since one page typically asks for several views of the same selection; the variables are then evaluated in memory and only the result (a table, a set of chart points, a matrix) is returned. The complete 2015–2025 selection of 46,988 unit-years loads in about 1.3 s on a laptop, and each subsequent view over it is computed in about 0.1 s.

3.3 Database additions

Migration 0009 adds two tables. `traci_factors` holds one row per TRACI version, pollutant, medium, and impact category, with the TRACI substance name and CAS number, the factor, its unit, and the workbook column it was read from; a unique index prevents duplicates. It is loaded from a six-row extract of the EPA workbook (`data/traci/traci_2_2_air.csv`), whose source, download date, and checksum are recorded beside it. `models` holds one row per trained model: its name and kind, target, inputs, tuned settings, training, validation, and test years, metrics, the JSON artifact used for prediction, the path of the saved scikit-learn object, and the software versions and Git commit that produced it. The committed database contains six factors and three models. Column-level documentation is in `DATABASE_BREAKDOWN.md`.

3.4 Prediction without Python on the server

`ml/train.py` implements its own preprocessing rather than scikit-learn's pipeline objects, so that every step can be written to JSON: missing numeric inputs are replaced by the training median, three skewed inputs are log-transformed ($\log(1+x)$), numeric inputs are standardized with the training mean and standard deviation, the fuel and unit type are one-hot encoded over categories seen at least 30 times, and the control flags are passed through. Ridge stores its coefficients and intercept; the tree stores its node arrays (children, split feature, threshold, and leaf value). `src/lib/model.ts` reimplements the same steps. scikit-learn compares tree features as 32-bit floats, so the TypeScript version rounds each feature to single precision before comparing it with a threshold; without this, values that fall between a threshold's single- and

double-precision forms take the wrong branch. A test replays 200 test-year unit-years exported by the training script and requires each model's TypeScript prediction to equal scikit-learn's within 10^{-6} (Section 6).

4 Methods

4.1 Variables and missing values

Table 3 defines the derived variables. A value is *missing* when CAMPD did not report it, or when a derived value's input is missing or its denominator is zero or negative. Missing values are excluded from statistics and counted separately; they are never treated as zero, which would bias every mean and intensity downward.

Table 3: Derived variables (computed per unit-year).

Variable	Unit	Definition
CO_2, SO_2, NO_x intensity	lb/MWh	mass (short tons) × 2,000 ÷ gross load
NO_x emission rate	lb/MMBtu	NO_x mass × 2,000 ÷ heat input
Heat rate	MMBtu/MWh	heat input ÷ gross load
Capacity factor	%	gross load ÷ (nameplate MW × hours in the year) × 100; 8,784 hours in leap years; values above 100 % are kept
Change (load, CO_2, SO_2, NO_x)	MWh or tons	this year minus the same unit's previous year
Change (%)	%	(this year − previous year) ÷ previous year × 100
TRACI impact, category c	kg eq	$\sum_i \text{mass}_{i,\text{kg}} \times \text{CF}_{i,c}$ (Section 4.4)
TRACI impact per MWh	kg eq/MWh	impact ÷ gross load

Combining groups. Each variable has one rule for combining unit-years into a group value. Masses, energy, changes in tons, and impacts are summed. Operating time and nameplate capacity are averaged per unit-year. Rates are *pooled*: the sum of the numerators divided by the sum of the denominators over the unit-years where the rate is defined. A state's CO_2 intensity is therefore its total CO_2 divided by its total gross load, not the mean of its units' intensities, which would weight a peaking turbine that ran ten hours as heavily as a baseload coal unit.

4.2 Descriptive statistics and correlation

Quantiles use linear interpolation between order statistics, $Q(p) = x_{(\lfloor h \rfloor)} + (h - \lfloor h \rfloor)(x_{(\lfloor h \rfloor+1)} - x_{(\lfloor h \rfloor)})$ with $h = (n-1)p$, which is Hyndman and Fan's type 7 definition [4] and NumPy's default. The standard deviation is the sample standard deviation $(n - 1)$. Pearson's r and Spearman's ρ (Pearson's r of average ranks) use the unit-years where both variables are reported.

4.3 Comparisons and tests

The 95 % confidence interval of a group mean is $\bar{x} \pm t_{0.975,\,n-1}\, s/\sqrt{n}$. Two groups are compared with Welch's t-test [11], which does not assume equal variances, and with the Mann–Whitney U test [6] (normal approximation with tie correction), which asks whether one group tends to have larger values and is robust to the heavy tails of emissions data. Three or more groups are

compared with one-way ANOVA and the Kruskal–Wallis test [5]. The distribution functions (t, F, χ^2, normal) are computed in TypeScript from the regularized incomplete beta and gamma functions and checked against SciPy [10].

Two assumptions limit these tests and are stated on the page. Groups differ in more than the attribute compared (size, age, fuel, hours run), so a difference is an association, not a causal effect of that attribute. And the tests treat every unit-year as independent, whereas the same unit appears in several years, so the p-values are optimistic.

4.4 TRACI impact assessment

CAMPD supplies the inventory, the mass of each pollutant a unit emitted in a year, and TRACI supplies characterization factors [1, 9], the potential contribution of one kilogram of a substance emitted to air to an impact category. The two are joined by an explicit mapping from CAMPD pollutant to TRACI substance; names are never matched approximately. CAMPD's CO_2 maps to *CARBON DIOXIDE* (CAS 124-38-9), SO_2 to *SULFUR DIOXIDE* (CAS 7446-09-5), and NO_x to *NITROGEN OXIDES*, for which TRACI lists no CAS number. TRACI also lists *NITROGEN DIOXIDE*, with a smog factor of 16.85 kg O_3 eq/kg against 24.79 for nitrogen oxides; CAMPD reports NO_x, so the nitrogen-oxides row is the correct one. Table 4 lists every non-empty air factor of the three substances in the TRACI 2.2 workbook.

Table 4: TRACI 2.2 factors used (per kg emitted to air).

Category	**CO_2**	**SO_2**	**NO_x**
Global warming (kg CO_2 eq)	1	—	—
Acidification (kg SO_2 eq)	—	1	0.7
Human health, particulate (kg $PM_{2.5}$ eq)	—	0.061111	0.0072222
Smog formation (kg O_3 eq)	—	—	24.793590

For each unit-year, CAMPD masses are converted from short tons to kilograms and multiplied by their factors, summing only within a category:

$$\text{Mass}_{i,\text{kg}} = \text{Mass}_{i,\text{short ton}} \times 907.18474, \qquad \text{Impact}_c = \sum_i \text{Mass}_{i,\text{kg}} \times \text{CF}_{i,c}, \qquad \text{Impact intensity}_c = \frac{\text{Impact}_c}{\text{Gross load}_{\text{MW}}} \tag{1}$$

Categories keep their own units and are never added together. If a pollutant that contributes to a category was not reported, that category is missing for the unit-year, while categories that do not need the pollutant are still computed: a unit-year without SO_2 has no acidification result but has a smog result. Group results are sums of unit-year results, and group intensities are pooled as above.

Worked example. In 2024 the four units of Ghent (facility 1356, Kentucky) reported 11,399,877.916 short tons of CO_2, 9,157.169 tons of SO_2, and 6,987.671 tons of NO_x, and a gross load of 11,580,029.2 MWh. Acidification is $(9{,}157.169 \times 1 + 6{,}987.671 \times 0.7) \times 907.18474 = 14{,}048.539 \times 907.18474 = 12{,}744{,}620$ kg SO_2 eq, or 1.1006 kg SO_2 eq per MWh. Global warming is $11{,}399{,}877.916 \times 907.18474 = 1.0342 \times 10^{10}$ kg CO_2 eq; particulate human health is 553,447 kg $PM_{2.5}$ eq; smog formation is 1.5717×10^8 kg O_3 eq. The Impacts tab, grouped by facility for 2024, shows the same values.

4.5 Prediction

Target. The model predicts a unit's annual NO_x mass in short tons. NO_x was chosen over CO_2 because CO_2 is almost a fixed multiple of heat input for a given fuel, so a CO_2 model would mostly recover that multiple; NO_x depends on combustion conditions and controls, so the controls carry information and every scenario of the specification (changed load, changed heat input, changed control) affects the prediction.

Inputs. Heat input, gross load, and operating time of the same year; nameplate capacity; unit age (reporting year minus the year of commercial operation); the unit's NO_x rate (lb/MMBtu) and operating time in the previous year; primary fuel and unit type; and five NO_x control flags (SCR, SNCR, low-NO_x burners, water, steam, or ammonia injection, and overfire air), read from the controls reported that year.

Leakage. No input is the target or computed from it: same-year NO_x mass, NO_x rate, and NO_x intensity are excluded. The previous year's NO_x rate is allowed because it is known before the year begins. Same-year heat input, gross load, and operating time are not known in advance, but they are the quantities the scenarios change, so they are kept; to show how much of the accuracy they provide, every model is also fitted without them (the ablation in Table 5).

Split. Unit-years from 2016 (the first year with a previous year) to 2025 with a reported NO_x mass are split by year: 25,116 training unit-years (2016–2021), 7,970 validation unit-years (2022–2023), and 7,755 test unit-years (2024–2025). Settings are chosen on the validation years; the chosen models are then refit on 2016–2023 and evaluated once on 2024–2025. No random split is used, since a random split would let a model learn a unit's 2024 emissions from its 2025 record.

Models. The baseline predicts the median NO_x of training unit-years with the same primary fuel and unit type (the overall median for unseen combinations). Ridge regression [3] is fitted to $\log(1 + \mathrm{NO}_x)$, because NO_x spans six orders of magnitude, with the penalty α chosen from $\{0.01, 0.1, 1, 10, 100\}$; its predictions are transformed back and floored at zero. The decision tree [2] is fitted to tons, with the maximum depth chosen from $\{4, 6, 8, 10, 12, 16\}$ and the minimum leaf size from $\{5, 20, 50\}$. Validation selected $\alpha = 10$ and a tree of depth 16 with at least 20 unit-years per leaf.

Metrics. Mean absolute error (MAE), root mean squared error (RMSE), the coefficient of determination (R^2), and mean absolute percentage error (MAPE). MAPE is computed only over unit-years of at least one ton, because peaking turbines that emit a fraction of a ton make percentage errors meaningless.

Sensitivity and scenarios. A sensitivity curve changes one input of the baseline unit-year by a percentage at each step, holds every other input fixed, and records the prediction, its absolute change, and its percentage change, $(\hat{y}_{\text{scenario}} - \hat{y}_{\text{baseline}})/\hat{y}_{\text{baseline}} \times 100$. An input is *outside the training data* when it lies below the minimum or above the maximum of that input in the training years, or when a fuel or unit type was not seen in training. These checks are per input; a combination of individually ordinary values can still be unlike any training unit-year.

5 Results

All results are for the reporting years 2015–2025 of the committed database unless stated otherwise.

5.1 Q1: emissions trends and the shift from coal to gas

Total CO_2 from the selected units fell from 2,098 million short tons in 2015 to 1,635 million in 2025 (22 %), and the pooled CO_2 intensity fell from 1,557 to 1,279 lb/MWh (18 %). Figure 4 shows where the decline came from: coal units emitted 1,393 million tons in 2015 and 739 million in 2024, while gas units rose from 688 to 824 million, and the two lines crossed in 2023, the first year in the data in which gas units emitted more CO_2 than coal units. Coal rose again in 2025 (825 against 799 million). Meanwhile the intensity of each fuel barely moved: coal units averaged 2,041 lb/MWh in 2015 and 2,036 in 2025, gas units 1,051 and 926. The sector's falling intensity is therefore mainly a change in the mix of generation, not cleaner coal plants.

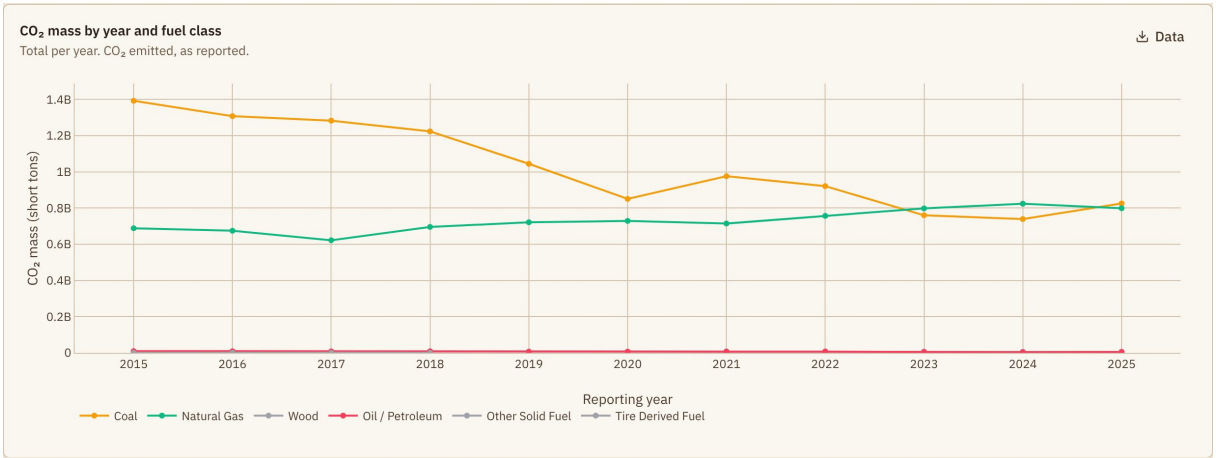


Figure 4: Trend chart: total CO_2 per year from coal and natural-gas units, 2015–2025.

5.2 Q2: NO_x rate and post-combustion control

Figure 5 shows the NO_x emission rate by fuel class on a logarithmic scale. Gas units have a median of 0.034 lb/MMBtu (interquartile range 0.013–0.096), coal units 0.136 (0.082–0.219), and oil units 0.574 (0.188–1.20), with wide overlap within each fuel.

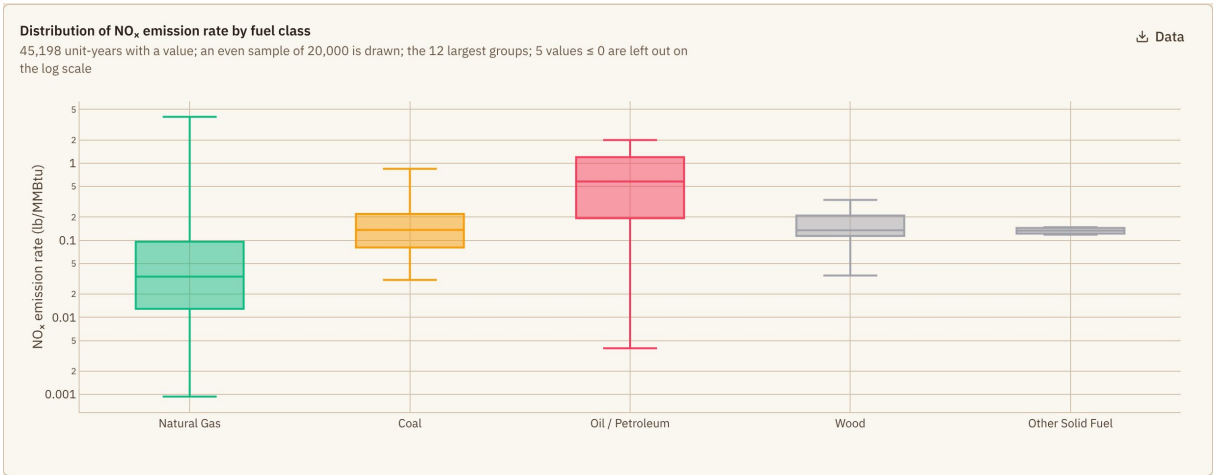


Figure 5: Box plot of the NO_x emission rate by fuel class (log scale).

For 2024, units whose NO_x controls include SCR had a pooled rate of 0.037 lb/MMBtu (1,547 unit-years) and units with no reported NO_x control 0.159 lb/MMBtu (479 unit-years), 330 % higher; the medians were 0.012 and 0.305 (Figure 6). Both Welch's test ($t = -22.3$, $p < 0.001$) and the Mann–Whitney test ($z = -29.8$, $p < 0.001$) reject equal groups. The difference is real but cannot all be attributed to SCR, because the two groups are different fleets: 881 of the 1,564 SCR unit-years of 2024 are gas combined-cycle units, while the 571 uncontrolled unit-years are mostly gas combustion turbines and oil units (326 gas and 212 oil). Comparing within one fuel and unit type, which the Compare tab supports by adding those filters, is the fairer reading.

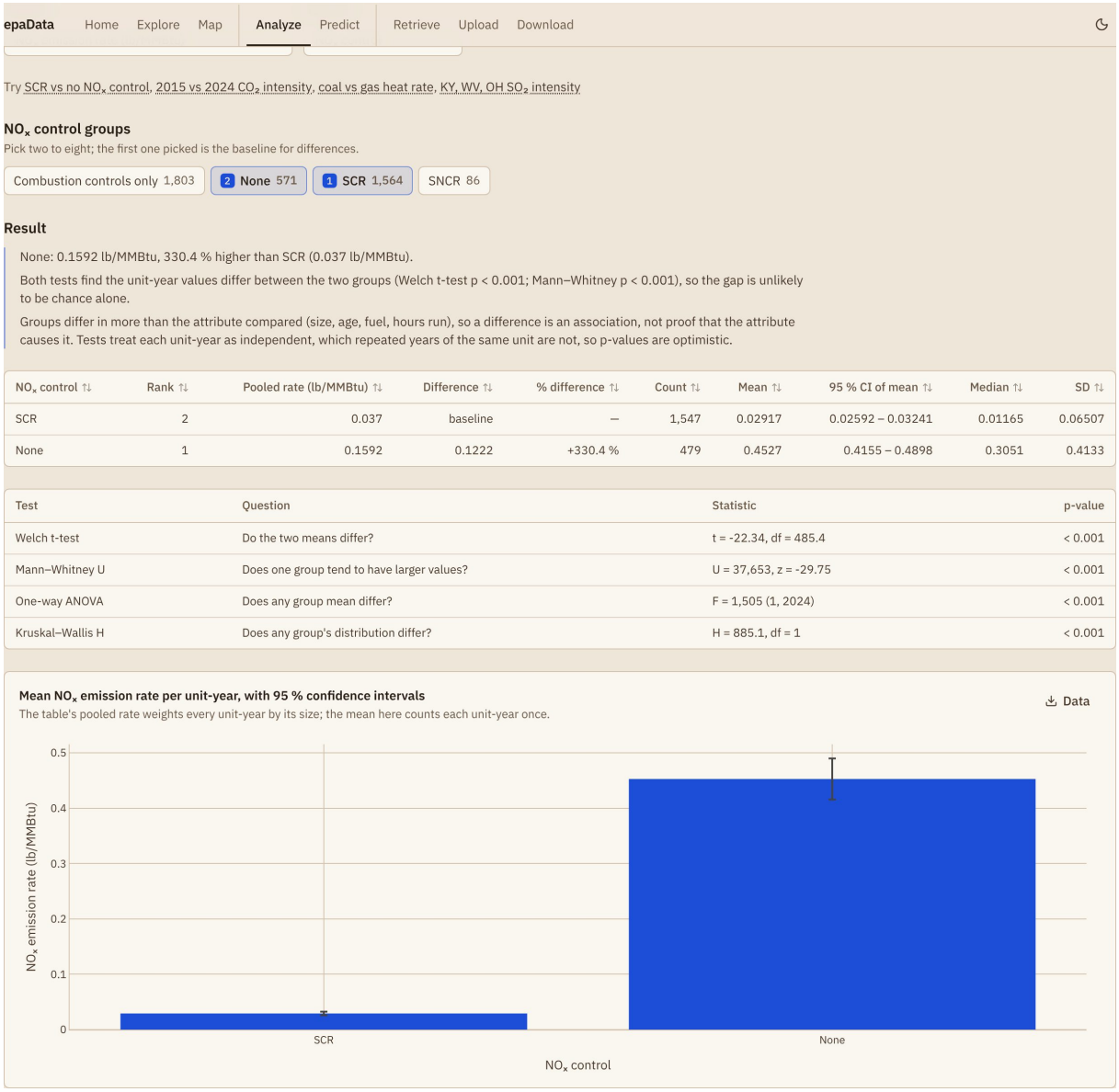


Figure 6: Compare tab: NO_x emission rate of units with SCR and with no NO_x control, 2024.

Unusual observations. Over 2015–2025, 3,582 unit-years lie above the far-out fence for NO_x intensity (6.1 lb/MWh, against a median of 0.58). Of these, 3,491 are combustion turbines, with a median operating time of 44 hours a year: peaking units whose start-ups produce NO_x with almost no generation. They are real reports rather than errors, but they are why the pooled rate (0.70 lb/MWh) is the honest summary and the mean (2.46) is not.

A misleading correlation. The Pearson correlation between CO_2 intensity and heat rate over the 37,718 unit-years reporting both is 0.9998, while Spearman's rank correlation is 0.765 (Figure 7). A few unit-years with tiny gross loads have intensities and heat rates thousands of times the norm; they dominate Pearson's r and nearly nothing else. The rank correlation is the honest summary for these heavy-tailed variables, which is why the chart offers both.

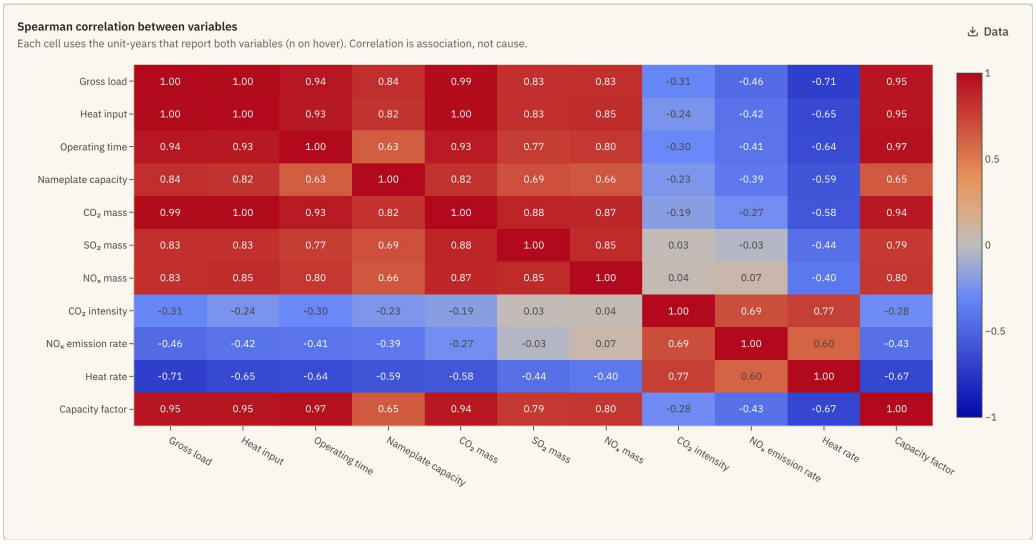
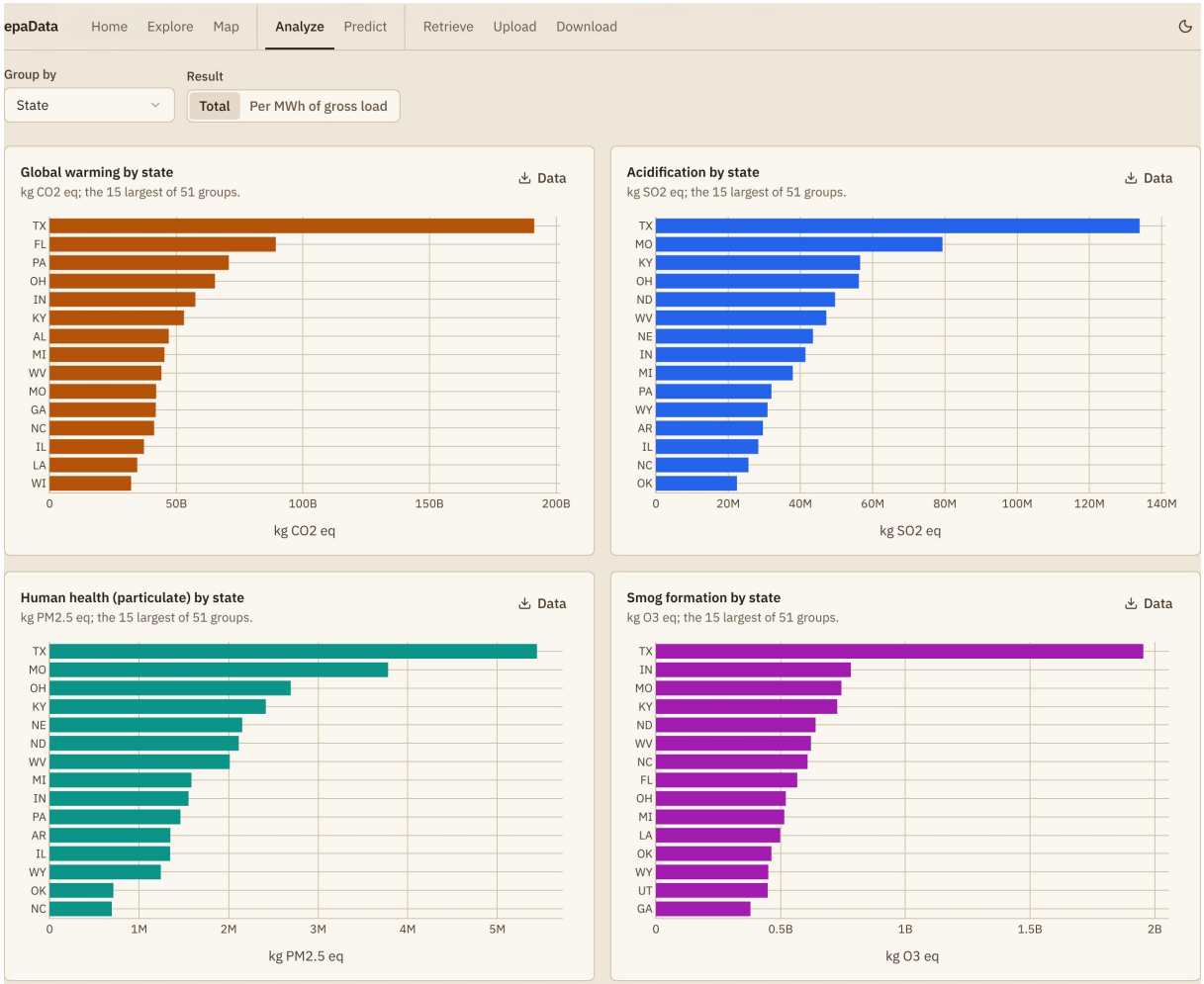


Figure 7: Spearman correlation heat map of eleven variables, 2015–2025.

5.3 Q3: impacts by state

In 2024 Texas contributed the most to every category: 1.91×10^{11} kg CO_2 eq of global warming and 1.34×10^{8} kg SO_2 eq of acidification, followed for acidification by Missouri (7.93×10^{7}) and Kentucky (5.65×10^{7}) (Figure 8). Per MWh the ranking changes: Nebraska (2.32 kg SO_2 eq/MWh), North Dakota (1.83), and Missouri (1.57) lead. Coal units produced 93 % of the reported gross load of Nebraska and North Dakota and 83 % of Missouri's, against 20 % in Texas, whose pooled SO_2 intensity was 0.51 lb/MWh against Nebraska's 3.95. Totals answer where impacts occur; intensities answer how clean a state's generation is.



Results by group

Each category is summed only over its own pollutants and keeps its own unit; categories are never added together. A unit-year is left out of a category when one of its pollutants was not reported. 51 of 51 groups, largest global warming first.

Download

State	Unit-years	Global warming (kg CO2 eq)	Acidification (kg SO2 eq)	Human health (particulate) (kg PM2.5 eq)	Smog formation (kg O3 eq)
TX	427	191,324,345,642	133,879,480	5,443,821	1,954,200,210
FL	208	89,296,594,211	22,359,299	553,993	566,508,819
PA	170	70,739,554,616	31,971,151	1,461,345	372,954,337
OH	120	65,271,002,975	56,171,729	2,693,391	520,716,790
IN	134	57,541,773,703	41,391,842	1,551,416	781,601,563
KY	90	53,055,832,472	56,527,753	2,414,705	726,826,533
AL	88	47,030,297,438	10,186,628	226,714	283,891,045
MI	118	45,341,819,903	37,876,668	1,583,997	514,774,689
WV	54	44,123,761,715	47,163,258	2,012,002	621,384,075
MO	103	42,053,981,802	79,309,808	3,780,363	743,584,808
GA	131	41,921,251,627	19,172,694	628,138	379,011,621
NC	125	41,237,122,332	25,612,831	694,747	608,056,313
IL	216	37,241,860,922	28,333,560	1,346,349	322,038,885
LA	79	34,603,150,368	22,115,817	637,124	498,163,965
WI	89	32,174,446,066	9,788,609	276,086	224,611,514
AZ	92	32,158,488,161	11,823,331	352,566	257,988,071
WY	25	30,103,432,916	30,893,785	1,241,802	450,574,121
CA	240	29,917,376,308	1,553,669	23,707	49,676,573
NY	256	28,311,266,734	5,039,215	84,154	174,485,004
VA	110	27,937,042,115	5,490,193	89,368	207,016,905
MS	60	27,697,600,812	8,357,787	174,287	237,780,772
OK	94	27,577,684,756	22,431,436	712,046	463,031,271
SC	61	27,162,814,257	15,431,954	565,273	264,059,435
ND	21	26,990,383,127	49,584,842	2,112,108	640,193,113
AR	41	26,625,231,068	29,609,390	1,348,564	321,393,894
CO	70	26,403,978,562	14,422,363	486,618	275,265,694
TN	98	24,519,465,134	13,818,244	577,050	226,722,512

5.4 Q4: prediction

Table 5 compares the models. Both trained models beat the baseline by a wide margin on the test years: Ridge reduces the MAE by 47 % and the tree by 76 %. The tree's validation and test results are close, and so are its training results ($R^2 = 0.956$), so it does not overfit noticeably despite its depth. Without same-year heat input, gross load, and operating time, validation R^2 falls from 0.853 to 0.704 for Ridge and from 0.967 to 0.837 for the tree: most of the remaining accuracy comes from the previous year's NO_x rate, which encodes the unit's combustion technology and controls.

Table 5: NO_x models: validation (2022–2023) and test (2024–2025) results.

	Validation		**Test**				**Ablation**
Model	MAE (t)	R^2	MAE (t)	RMSE (t)	R^2	MAPE	val. R^2
Baseline (median by fuel and type)	114.2	0.474	109.4	366.9	0.489	143.6 %	0.474
Ridge ($\alpha = 10$, log target)	57.9	0.853	58.5	213.5	0.827	49.3 %	0.704
Decision tree (depth 16, leaf 20)	26.5	0.967	26.3	113.9	0.951	20.9 %	0.837

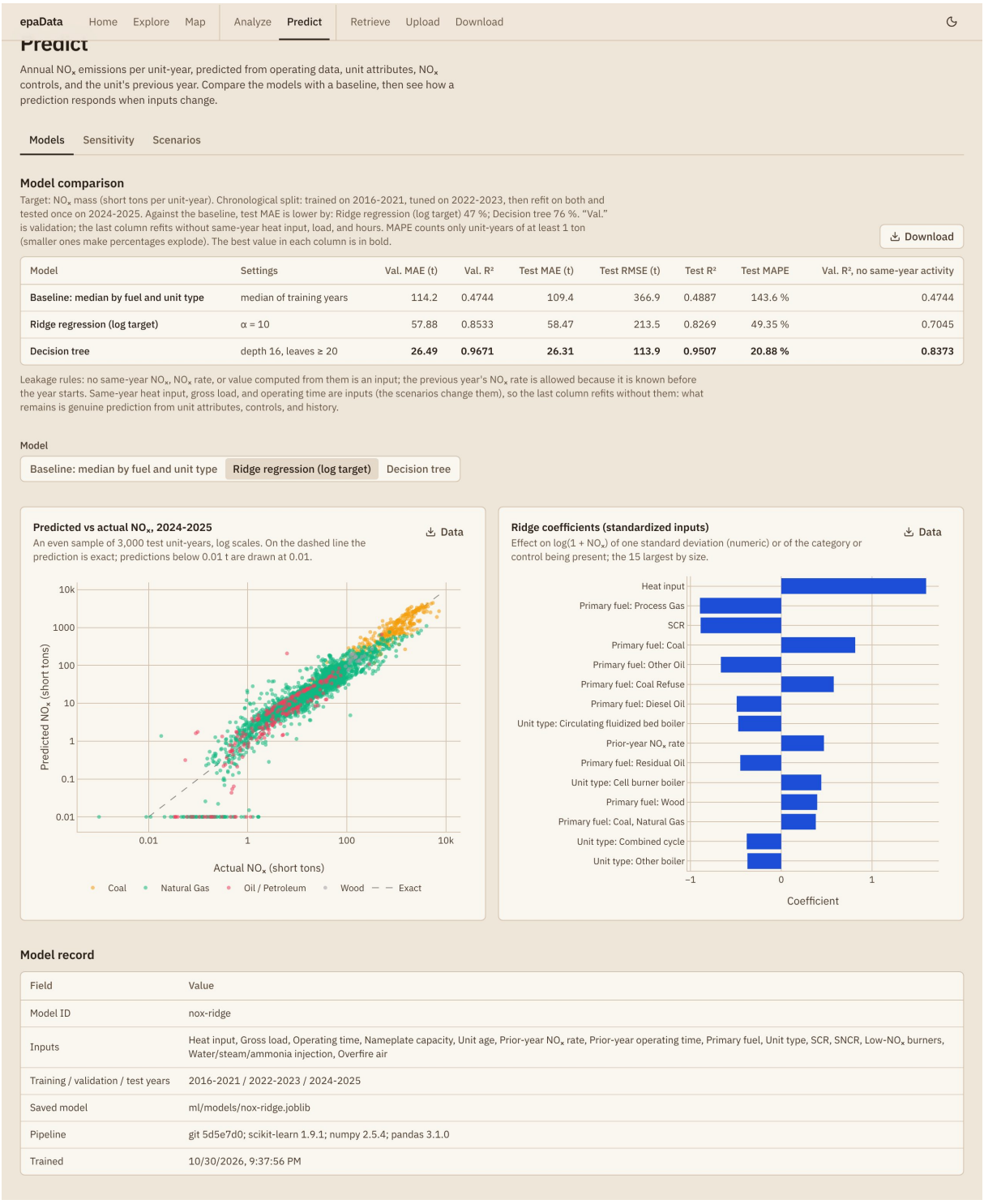


Figure 9: Models tab: comparison with the baseline, predicted against actual NO_x for the test years, and the Ridge coefficients.

Sensitivity. Figure 10 varies the heat input of Ghent unit 2 in 2024, a 556 MW tangentially fired coal unit with low-NO_x burners and overfire air but no SCR, which emitted 4,146 tons of NO_x. Ridge predicts 2,972 tons at the baseline; a 50 % increase in heat input raises the prediction to 3,827 tons (29 %), an elasticity of about 0.6. Beyond 85.8 million MMBtu, the largest heat input in the training years, the curve enters the shaded region and its points turn amber.

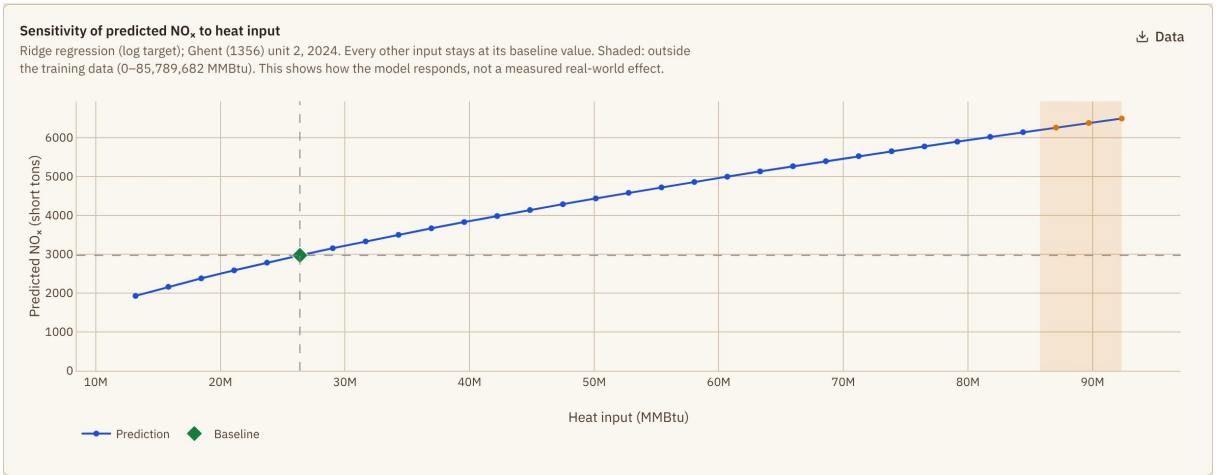


Figure 10: Sensitivity of Ridge's NO_x prediction to heat input for Ghent unit 2, 2024, from −50 % to +250 %. Shaded: outside the training data.

Scenarios. Figure 11 applies the named scenarios to the same unit. The two models agree on heat input (−5 %: Ridge −3.2 %, tree −9.1 %) and on gross load (nearly no effect once heat input is fixed), but disagree on controls and fuel. Ridge predicts that adding SCR would cut NO_x by 59 % and a switch to pipeline gas by 69 %; the tree predicts no change for either. The tree splits mostly on heat input and the previous year's NO_x rate, which still describes the unit as it was, so a changed control flag rarely changes its path; Ridge applies one average SCR coefficient to every unit. Neither number is a causal estimate: where two reasonable models disagree, the data do not settle the effect, and the page says so.

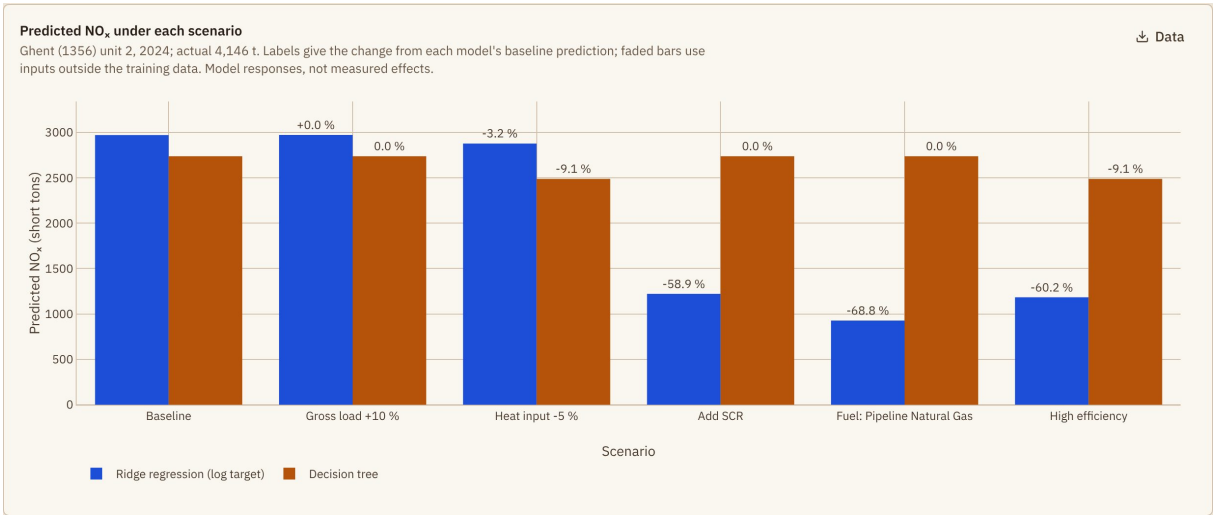


Figure 11: Scenarios for Ghent unit 2, 2024, under Ridge and the decision tree; labels give the change from each model's baseline prediction.

6 Testing and results

6.1 Approach

Each calculation was tested against an independent reference: statistics against NumPy and SciPy, TRACI results against hand calculation, and TypeScript predictions against scikit-learn. Every page and chart was then exercised in a headless browser (Playwright with Chromium) on

the development and production builds, at desktop and phone widths, checking for console errors and reading each screenshot; downloaded ZIP files were unpacked and their CSV and metadata checked. `npm run validate` (formatting, linting, type checking, unit tests, and a production build) passes.

6.2 Unit tests

`npm test` runs 60 TypeScript tests (one Phase 1 test is skipped), of which these are new:

- `stats.test.ts`: quantiles and standard deviation against NumPy; Pearson and Spearman correlation, Welch's t, Mann–Whitney U, the confidence interval, ANOVA, Kruskal–Wallis, and the χ^2 tail against SciPy, to 10^{-9} for statistics and 10^{-6} for p-values; pairwise-complete correlation; constant inputs.
- `analysis.test.ts`: each derived rate, and its missing value when an input is missing or a denominator is zero; capacity factor in leap and ordinary years; year-over-year changes with a missing or zero previous year; pooled rates as a ratio of sums; TRACI impacts against hand calculation, with categories kept apart and a missing pollutant propagating only to its categories; the factor extract (six factors, nitrogen oxides rather than nitrogen dioxide); the control classes.
- `model.test.ts`: the parity test (all three models on 200 test unit-years, within 10^{-6}); input derivation (unit age, control flags, a previous rate that is missing when the previous heat input is zero); out-of-range detection; the direction of the response to heat input; readable feature names.
- The URL state: the default year range applies whenever no filter is in the URL; signed percentage formatting.

`npm run test:py` runs six pytest tests of the training pipeline: the three periods do not overlap; no input is the target or derived from it; the input rules; the preprocessing specification produces the same matrix after a round trip through JSON; the baseline is the group median; and MAPE ignores targets below one ton.

6.3 Defects found and corrected

Testing found several defects, all corrected. The 60-second row cache returned a Drizzle query object rather than its result; Drizzle queries are lazy and re-run on every `await`, so the "cached" selection was queried again for every view (1.5 s each); the loader now returns a settled promise and a cached view takes under a millisecond. The server export imported the module that defined the browser-side export context, which Next.js rejects in server code, so the dataset export failed; the context moved to the client component that uses it. The year-range chip, shared with the Explore page, displayed "2,015–2,025" (a Phase 1 defect); years are now shown without separators. A chart's loading placeholder rendered its spinner and text side by side; the Predict page overflowed a phone screen by ten pixels; both were fixed. The Python requirements had been pinned to versions that need Python 3.11 or later, while the documented setup uses Python 3.9; they are now minimum versions that support both.

A final audit against the specification found five more. The emissions-program filter that §4.1 lists had never been built, although the filter bar offered controls; it now matches a whole program code, since CAMPD stores lists such as "ARP, CSNOX, CSOSG2" and a substring match would let "CSNOX" catch "CSNOXOS". The default 2015–2025 range applied only to a URL with no parameters at all, so a link to a tab (`/analyze?tab=charts`) silently analysed every year including the partial 2026; it now applies whenever the URL sets no filter. The row

cache did not notice an import for up to a minute; its key now includes the time of the latest import. A caption wrapped around a segmented control as an HTML `label` forwarded a click on the caption to the first button, flipping the choice; such captions are now plain groups. Finally, sorting groups whose value was missing used an inconsistent comparison.

6.4 Assessment against the objectives

- *Descriptive statistics, trends, relationships, unusual observations:* met by the Statistics and Charts tabs (Q1, the correlation outlier).
- *Comparisons:* met by the Compare tab with tests and plain-language readings (Q2).
- *Interactive visualization:* met; seven chart types with export.
- *TRACI impacts:* met for the four categories TRACI 2.2 defines for these pollutants (Q3).
- *Prediction:* met; two models beat the baseline on held-out years (Q4).
- *Sensitivity and scenarios:* met, with range warnings.
- *Saving and exporting:* met; every result exports with metadata.
- *Explaining methods and limitations:* met in the interface (definitions, captions, caveats) and in this report.

7 Discussion

7.1 Effectiveness of techniques and tools

Treating every derived indicator and TRACI category as an ordinary variable, one definition with its unit, aggregation rule, and missing-value rule, was the most effective decision of the phase. Statistics, charts, comparisons, impacts, and exports are each written once and work for all 29 variables; adding a variable means adding its definition. Reusing the Phase 1 filters for the analytical dataset meant that the dataset builder came with every filter, its URL encoding, and its tests already in place.

The offline-training design kept the strengths of both languages: pandas and scikit-learn for fitting and tuning, and TypeScript for interactive use. Its cost is that preprocessing is written twice, in `train.py` and `model.ts`; the parity test is what makes that safe, and it would catch any change to one that is not made to the other.

7.2 Technical choices and alternatives

Flask and pandas at request time. The specification recommends computing analyses with pandas behind Flask routes. Rewriting the Phase 1 application in Flask was out of the question, and calling Python from each request (as Phase 1 does for uploads) would have added a process start to every chart and would not run on the hosted build. The statistics needed are few and well defined, so implementing them in TypeScript and checking them against SciPy was the smaller and safer path.

Plotly.js. Plotly provides every chart type the specification lists, including box plots, heat maps, and state maps, together with tooltips, zoom, and PNG export, so a single component wraps it for all charts. Its bundle is large, so it is loaded only when a chart is first shown.

Ridge and a decision tree. The two models were chosen to differ: Ridge is smooth and gives every input a single global effect, which makes sensitivity curves easy to read; the tree captures interactions and thresholds, and is far more accurate, but its response to a single input is a step function and it may ignore an input entirely. The scenario disagreement in Section 5 is a direct consequence, and showing both models makes the uncertainty visible instead of hiding it behind the more accurate one.

7.3 Challenges and lessons learned

The TRACI workbook needed care: it lists both nitrogen oxides and nitrogen dioxide with different smog factors, and it has no eutrophication column, although the plan expected one. Reading the source rather than relying on a summary avoided both errors. A similar lesson came from the parity test: a TypeScript reimplementation of a scikit-learn tree that looks correct still differs in rare cases unless it reproduces the library's 32-bit comparison. Finally, the correlation outlier showed why statistics over raw emissions data must be robust by default: a handful of unit-years with almost no generation can produce a near-perfect Pearson correlation that describes nothing.

8 Limitations and future improvements

8.1 Current limitations

Potential, not actual, impacts. TRACI factors express potential contributions for comparison. They do not account for stack height, local population, weather, or background concentrations, and they say nothing about actual illness or damage. Only the three pollutants CAMPD reports are covered; particulate matter, mercury, and other emissions are not, and TRACI 2.2 provides no eutrophication factor for these substances.

Missing pollutants remove categories. A category is missing for a unit-year when one of its pollutants is not reported. About 13.7 % of 2015–2025 unit-years lack SO_2, so their acidification and particulate results are missing and group totals cover only the unit-years that report every contributing pollutant; the Dataset tab shows these counts.

Fuel and unit type are not versioned by year. As in Phase 1, fuel and unit type are stored once per unit, so a unit that switched from coal to gas is treated as a gas unit in every year, both in grouped statistics and as a model input.

Statistical independence. Unit-years of the same unit are correlated, but the tests treat them as independent, which overstates their significance. A mixed model or tests on one value per unit would be more rigorous.

Model responses are not causal. Both models learn associations in observational data. A scenario shows how a model responds to a changed input with everything else fixed, which may be physically impossible (for example more gross load with the same heat input). The previous year's NO_x rate, the most useful history input, also describes the unit's existing controls, which weakens the models' response to a changed control. Range warnings check each input separately, not the combination.

Other limitations. The state map loads the U.S. outlines from Plotly's content network, so it needs internet access. Analysis configurations are saved only as URLs. The year-over-year variables compare consecutive years only, so a unit with a missing year has no change for the following year.

8.2 Future improvements

Storing fuel and unit type per year, as Phase 1 already proposed, would correct both the grouped statistics and the model inputs. Tests clustered by unit would give honest p-values. Gradient-boosted trees with monotonic constraints, and models fitted separately for coal and gas, would likely improve accuracy and make the response to controls more plausible. Joint out-of-distribution checks (for example a nearest-neighbor distance to the training data) would make range warnings stricter. Finally, adding the TRACI factors for particulate matter and mercury would become worthwhile once CAMPD's mercury data are imported.

9 Individual contributions

The Phase 2 work kept the roles of Phase 1, a back-end and a front-end lead, with testing, documentation, the report, and the presentation shared.

Raaj Banga (back-end lead). Responsible for the analytical data layer and the models: the variable catalogue of derived indicators and year-over-year changes with their missing-value rules; the statistics and hypothesis tests and their checks against SciPy; the TRACI 2.2 extract, the `traci_factors` and `models` tables, and the impact calculation; the analysis and prediction procedures and the analytical-dataset export; the training pipeline in `ml/train.py`, the TypeScript inference module, and the parity and pipeline tests; and the README, `DATABASE_BREAKDOWN.md`, and this report.

Arnav Bawankule (front-end lead). Responsible for the user interface of Phase 2: the Analyze page and its Dataset, Statistics, Charts, Compare, and Impacts tabs; the Predict page and its Models, Sensitivity, and Scenarios tabs; the shared Plotly chart component, its theming, and the export controls that save each chart and table with its metadata; and the presentation slides.

10 References and acknowledgments

10.1 Data

Emissions, operating, facility, and unit data are those of the Phase 1 database, from the EPA Clean Air Markets Program Data (CAM API, `https://www.epa.gov/power-sector/cam-api-portal`), a U.S. government work in the public domain. TRACI 2.2 characterization factors are from the EPA workbook `traci_2_2.xlsx` [9], downloaded on 30 October 2026 (SHA-256 `80383dec…3725334`); only the six factors in Table 4 are redistributed, in `data/traci/`.

10.2 Reused code and assets

Phase 2 builds on the Phase 1 code base and its libraries. It adds Plotly.js [8] for charts, fflate for ZIP files, and, in the offline pipeline, pandas, NumPy, scikit-learn, joblib, SciPy, and pytest,

all used under their open-source licences. No external example code was copied.

10.3 Generative AI disclosure

In accordance with the course policy, we disclose that generative AI was used throughout this phase. We used Anthropic's Claude, through the Claude Code tool, to help plan the phase, write and review application, pipeline, and test code, investigate defects, capture screenshots, and draft the README, the database documentation, and this report. The team defined the analytical questions, the prediction target, and the design decisions, reviewed all generated code and text, and accepted changes only after verifying them with the type checker, linter, unit tests, reference values from NumPy and SciPy, and manual use of the application. Every number in this report was produced by running the application or the training pipeline against the committed database. The team takes responsibility for the content of the code and of this report.

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We used AI to help with this project.